

Astronomy of Stars and Galaxies AST 102

AST102 CRN 95204

Class modality: in person, on campus

Meeting times: Tuesdays 12:30-1:45pm (T-211), Thursdays 12:30-1:45pm and 1:50-3:40pm (T-201)

Professor: Dr Glenda Denicoló

Email: denicog@sunysuffolk.edu (email is the best way to contact me)

Office hours (office T-218): Mon Wed 8:30-9:30am, Mon 10:45am-12:15pm, Tue 9:30-11am.

Tutoring: free tutoring is available in T-210. Check the schedule posted at the door of T-210, and in D2L.

Textbook: Astronomy by OpenStax (free digital copy) <https://openstax.org/details/books/astronomy-2e>

Required for Labs and HW: Registration link inside D2L. Cost: \$10/semester/student.

Other required materials: always bring a Scientific Calculator. No cell phones will be accepted as calculators.

Catalog Description

Introduction to fundamental aspects of universe beyond our solar system. Topics include properties of electromagnetic radiation and its relation to study of celestial objects; structure, classification and evolution of stars, nebulae, star clusters, galaxies, and material between stars. Age, origin and evolution of universe studied in terms of modern cosmology. Occasional evening observations required. Note: Fulfills SUNY-GE Natural Sciences. (3 hrs. lecture, 2 hrs. laboratory.) Prerequisite: MAT007 or equivalent. Offered on: A-E-G / 4 cr. hrs.

Course Learning Outcomes

Upon completion of this course, students will be able to:

1. Use measurements and astronomical data to describe the universe.
2. Summarize how the scientific method applies to astronomy.
3. Explain how the properties of light are used to obtain information about the universe.
4. Compare the fundamental properties of stars to the Sun and each other.
5. Describe the structure of the Sun.
6. Trace the evolution of stars from birth to death.
7. Compare the fundamental properties of galaxies to the Milky Way and each other.
8. Describe the past and future evolution the universe.
9. Discuss recent developments in the field of astronomy.

Grade Policy

Item	Grade weight
Labs (13 – 3 lowest = 10; or 3 points each) – no make-up!	30%
Homework (10 – 2 lowest = 8) – no make-up!	13%
Non-cumulative Tests (3; 15 points each) – no make-up!	45%
Optional cumulative Final Test (can replace lowest Test if taken)	(15%)
Planetarium Test – no make-up!	10%
Class attendance (> 90%, or 0-2 missed classes)	1%
Pre- and post-assessments together	1%
EXTRA: 5-min presentation at the end of the semester	2%
EXTRA: Attending an observing night or talk at SCCC or SBU (1)	1%
Total =	103% !!

The final course grade will be a combination of the 10 best lab and 8 best homework grades, 3 tests (if Final Test is taken, it can substitute the lowest grade of one of the first 3 tests), 1 planetarium test, class attendance, and completion of course assessments. Optionally, students can be awarded extra credits by giving a 5-min presentation at the end of the semester (see schedule), and by attending an observing night or talk (see information in D2L). No other form of extra credit will be given.

There is **no make-up for a missed Test, without exception**, because the lowest grade among all 4 tests is dropped and only 3 are used to compute the final grade. If a student misses one of the initial 3 tests, the students should take the cumulative Final Test to replace the missed one.

There are **no make-ups for missed Labs or Homework, without exception**, because several of the lowest grades (3 lowest for Labs, and 2 lowest for HW) are dropped. Once the deadline for submission has passed, students will be able to access solutions even if the activity was missed, in order to study for Tests.

Grades will always be posted in D2L. Student must check grades regularly. If there are any inconsistency or questions, **consult the instructor within a week.** Do not leave problems or questions to the last week of the semester. Issues must be reported immediately.

***Reach out and speak up as soon as possible when/if an issue comes up.
Communications of extenuating circumstances done only at the end of the semester will have no effect.***

89.5 ≤ A ≤ 100

84.5 ≤ B+ ≤ 89.4

79.5 ≤ B ≤ 84.4

There will be NO curving of the grades in this course (i.e., the curving is done by the student when completing extra credit assignments). Your final grade is non-negotiable. The letter grade breakdown that will be used is shown on the right.

74.5 ≤ C+ ≤ 79.4

69.5 ≤ C ≤ 74.4

64.5 ≤ D+ ≤ 69.4

According to the Family Educational Rights and Privacy Act (FERPA), grades will never be discussed via e-mail or phone, only in person. Always check your grades in D2L.

59.5 ≤ D ≤ 64.4

F ≤ 59.4

Lab activities and policies

There are a total of 13 lab activities that **must be completed while working in groups in class, and must be submitted in class** (this is not an online lab class, no remote lab work will be accepted). Everyone in a group will receive the same grade. However, *if a member of the group departs before the submission of the lab, that group member will be given zero for that lab activity.*

There is **no make-up for the labs** because the 3 lowest lab grades will be dropped at the end of the semester.

Links to lab activities are available inside D2L/Brightspace.

Students are welcome to bring their own computer device if they wish (**no tablets or cell phones** should be used, as they will *not* display the lab information correctly), but **school laptops will also be available**. No installation is necessary, as PivotInteractives **runs in a browser**. Preferred browsers are Chrome, Firefox, or Safari.

The cost to use PivotInteractives is \$10 for the entire semester per student (payment is required at registration). **The registration link is available inside D2L:** enter our astronomy course page, then click “Content” (top menu), and find the link inside “PivotInteractives REGISTRATION” (menu on the left).

Solutions and corrections will be available within a week of the submission of the lab. Lab grades will also be posted in D2L. **Students should consult the graded labs to study for Tests.**

Homework assignments and policies

Homework assignments are available inside Pivot Interactives like the Lab activities; there is no additional cost.

Homework assignments must be **completed INDIVIDUALLY** before each deadline (see detailed schedule). HW can be completed remotely outside of class time, and is *not* group work.

There is **no make-up for HW** because the 2 lowest grades will be dropped at the end of the semester.

Solutions and corrections will be available within a week of the submission of the HW. Grades will also be posted in D2L. **Students should consult the graded HW to study for Tests.**

Planetarium

There will be one Planetarium Test at the end of the semester worth 10% of the final grade.

Students are encouraged to access the planetarium room (**always complete the sign-in sheet at the door**) during the semester to practice for this test. In D2L, students can find the explanation of the remote-control used in the planetarium, and the list of constellations, asterisms and stars that will be asked in the Planetarium Test.

If the planetarium room is locked, students can ask access to any astronomy faculty available (my office is 218), or to Mrs. Deanna Downs (office 220), or book a time in advance.

During the semester, the instructor will visit the planetarium a few times with the students to guide them through the material that will be asked in the Planetarium Test.

Our attendance policy

(also check the College Policies included in this document)

Due to the nature and amount of the material in this course, it is crucial that each student attend every lecture and laboratory session.

In our course, students will be awarded 1 point in their final grade if they miss only 0-2 classes per semester, which corresponds to more than 90% attendance. Anyone who misses more than 2 classes will not be awarded the attendance point, without any exception, even if there is a valid excuse. In that case, the

student must take advantage of other opportunities (such as extra credit assignments) to make-up for the missed attendance point.

Students missing classes must contact the instructor as soon as possible to learn about any updates or announcements that may have happened during the missed class. **It is the student's responsibility to reach out as soon as possible.**

Withdrawal policy

(also check the College Policies included in this document)

Students can withdraw by completing a form and submitting it to the registrar's office by **Monday November 10, 2025** (deadline). It is common *courtesy to previously communicate to your instructor that you intend to leave the course*. <https://www.sunysuffolk.edu/current-students/registrar/withdrawal.jsp>

Only students who submit a course withdrawal form on or before the deadline are guaranteed a W.

Withdrawals after the deadline can only be done in extreme cases such as medical leave or extended jury duty, and never along the lines "I am failing the class, thus I wish to withdraw at this late stage". W requests after the deadline are judged by a college dean, not by the instructor.

Please note that **if you stop attending class without officially withdrawing, you will receive an F at the end of the semester**. It is important to know also that the instructor has no means to give you a W at the end of the semester (this option is not available in the school system when entering the final grades). **Withdrawal is an action started by the student** and it must go through the registrar's office before the W deadline.

College Policies

(TBD ~1 week before classes begin)

AST102 CRN 95204 Tue 12:30-1:45 (T-211), Thu 12:30-1:45, 1:50-3:40 (T-201)

W	Date	HW # due	Activity	Textbook Chapter/Section
1	Tue Sep 2		Introduction to AST102; <i>pre-assessment to evaluate the course</i>	
	Thu Sep 4		Brief overview of astronomy Lab 1: Planetarium with https://stellarium-web.org/ ; Planetarium room	1
2	Tue Sep 9		Scales, the universe in powers of ten, units, constellations, horizon system	2.1, 4.1, App C, D
	Thu Sep 11	1	Patterns and cycles in the sky, seasons Lab 2: Overview of mathematical and scientific tools	4.1-4.2, App C, D
3	Tue Sep 16		Basic properties of light, electromagnetic spectrum	5
	Thu Sep 18	2	Atomic transitions, types of astronomical spectra Lab 3: Basic coordinates and seasons	5
4	Tue Sep 23		Doppler effect, astronomical instruments	5, 6
	Thu Sep 25		Newton's laws, Law of gravitation Lab 4: Gravity and orbits – Part 1	3
5	Tue Sep 30	3	Test 1 (1:15 min)	Study slides in D2L
	Thu Oct 2		The Sun (structure, composition, solar cycle) Lab 5: Gravity and orbits – Part 2	15
6	Tue Oct 7		The Sun (nuclear fusion)	16
	Thu Oct 9	4	Analyzing star light (brightness, color, spectra) Lab 6: The Sun	17
7	Tue Oct 14		Stellar spectra	17
	Thu Oct 16	5	Surveying stars (mass, diameter) Lab 7: What is this star made of?	18
8	Tue Oct 21		Surveying stars (variable stars, star clusters)	19, 22
	Thu Oct 23	6	Surveying stars (formation of stars) Lab 8: The HR diagram – Part 1	20, 21
9	Tue Oct 28		Life cycle of stars (cycle for low-mass stars)	22
	Thu Oct 30		Life cycle of stars (cycle for high-mass stars) Lab 9: The HR diagram – Part 2	22
10	Tue Nov 4	7	Test 2 (1:15 min)	Study slides in D2L
	Thu Nov 6		Stellar graveyard (white dwarfs, neutron stars, pulsars, binary systems, blackholes, gamma-ray bursts) Lab 10: Distance to the stars (brightness and luminosity)	23, 24
11	Tue Nov 11		No class – Veteran's day	
	Thu Nov 13		The Milky Way (characteristics) Lab 11: Cosmic distance ladder	25
12	Tue Nov 18		The Milky Way (stellar populations, formation, its center)	25
	Thu Nov 20	8	Galaxy clusters; distance measurements Lab 12: Expanding universe; Planetarium room	26
13	Tue Nov 25		Galaxy evolution, Quasars, Active Galactic Nuclei (AGNs)	28
	Thu Nov 27		No class – Thanksgiving recess	
14	Tue Dec 2		Evidence for dark matter	27
	Thu Dec 4	9	Dark matter and large-scale structure formation; dark energy Lab 13: Doppler effect	28, 29
15	Tue Dec 9		The Big Bang; Cosmic Microwave Background (CMB) radiation	29
	Thu Dec 11		<i>Post-assessment to evaluate the course.</i> Extra credit presentations; Planetarium Test	
16	Tue Dec 16	10	Test 3 (1:15 min)	Study slides in D2L
	Thu Dec 18		Optional cumulative Final Test (1:15 min)	

The current schedule is subject to changes due to any class cancellations or adjustments. Students must attend class to learn about any changes, or must email instructor immediately to learn, in case a class is missed.

DISCLAIMER

General rules are posted in your course outline. But even if a rule is not specifically shown in your course outline, it may be announced only in class and it is a valid new rule that must be followed. If there are changes to any existing rules, these too will be announced in class whether or not they are in print.

Students are responsible for apprising themselves of anything that transpires in class whether or not they are in attendance. All students need to be aware of all announcements whether or not they are in attendance: 1) ask another student; 2) e-mail the professor and ask to update you on any new announcements. If you missed a class, you must ask: it is YOUR responsibility to ask.

Professors are not “babysitters”, i.e., they will not remind you all the time of the requirements presented in the course outline: you have the responsibility to know what is strictly expected from you. If you miss a deadline, it’s on you. Print/take a photo of the detailed schedule and have it handy at all times.

All students are personally accountable for all submitted work. You must keep track of your own progress in class. Learn where to check your grades and your calculated final grade at all times (it’s in the gradebook in D2L!); estimate your own final grade to know where you are heading. This only requires basic knowledge of math: ask for help if needed. Ask for help at the start of the semester, do not let problems linger.

YOU are responsible for the grade you receive.

Professors are just messengers in the classroom: the grade you get is not “given” by the professor – it is the grade that you gave to yourself, the grade that you have earned, in direct correspondence to how much YOU have studied, worked and learned about the subject. Taking responsibility for your own acts and taking responsibility for your own work is a good rule for life.

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Return this signed slip to the professor by the 2nd class meeting.

I have read and am aware of all the information contained in the course outline of AST102.

Date: ____/____/____

Name (please print): _____

Your signature: _____